

DO STUDENTS WHO FEEL SAD HAVE DIFFERENT SCHOOL SAFETY INDEX SCORES IN DIFFERENT GRADES?

DATA USED

YRBS_2007 .csv
cleaned_school_safety.csv
high_risk_students.csv

VARIABLE USED

'Q4': 'Grade'
'Q14': 'Weapon_School'
'Q18': 'Fight_Total'
'Q20': 'Fight_School'
'Q23': 'Sad_Hopeless'
'Q56': 'Injected_Drug'
'Q57': 'Drug_School'

RESEARCH METHODS

Data cleaning : Clean missing values and contradictory issues

Safe score : used five variable to generate. Q14、Q18、Q20、Q56、Q57

Two-Way ANOVA : Analyze the main effects and interactions of grade level and the mental health and safety index.

Two-Way ANOVA

Source of Variation	Sum of Squares (SS)	Degrees of Freedom (df)	F-statistic (F)	p-value (P)
Grade Level (C(Grade))	5194.05	3.0	4.5642	0.0034 *
Mental Health (C(Sad_Hopeless))	133907.0	1.0	353.0105	< 0.001 **
Interaction (Grade × Mental Health)	977.523	3.0	0.8590	0.4616
Residual (Error)	5064795.0	13352.0	—	—

Result:

Mental Health and Grade are significant p=0.0000, p=0034
But Interaction is not significant p=0.4616

Grade (C(Grade))	Mental Health (C(Sad_Hopeless))	Sample Size (N)	Mean Risk Score (0-100)
1.0 (9th)	No (0)	2339	9.09
	Yes (1)	953	16.94
2.0 (10th)	No (0)	2328	9.48
	Yes (1)	994	15.66
3.0 (11th)	No (0)	2385	8.49
	Yes (1)	967	15.46
4.0 (12th)	No (0)	2362	7.87
	Yes (1)	1032	14.66

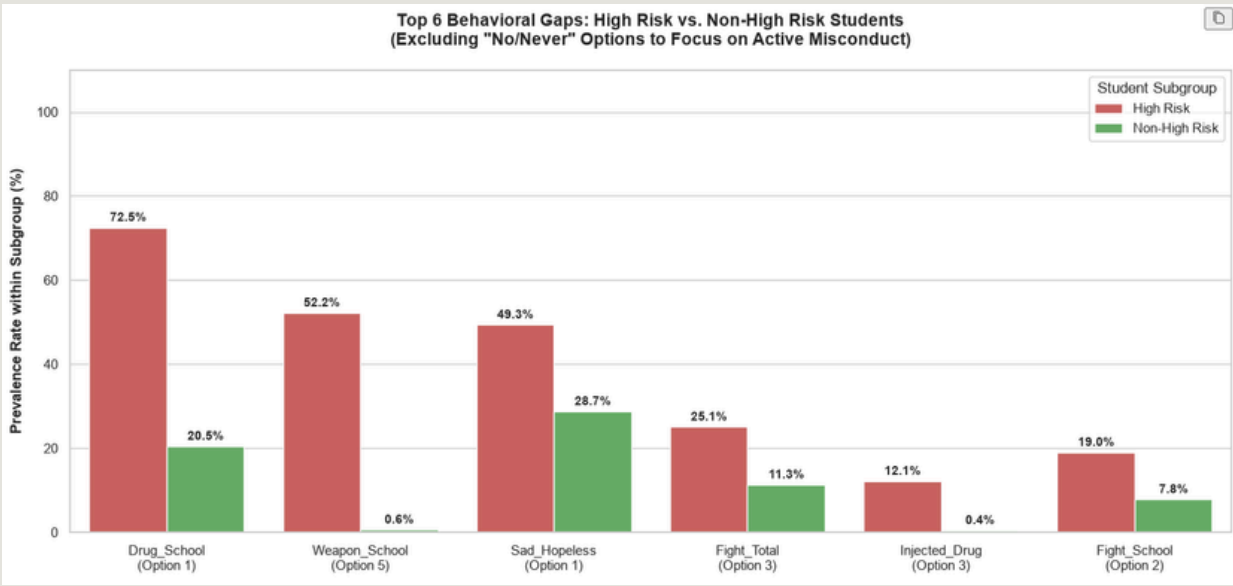
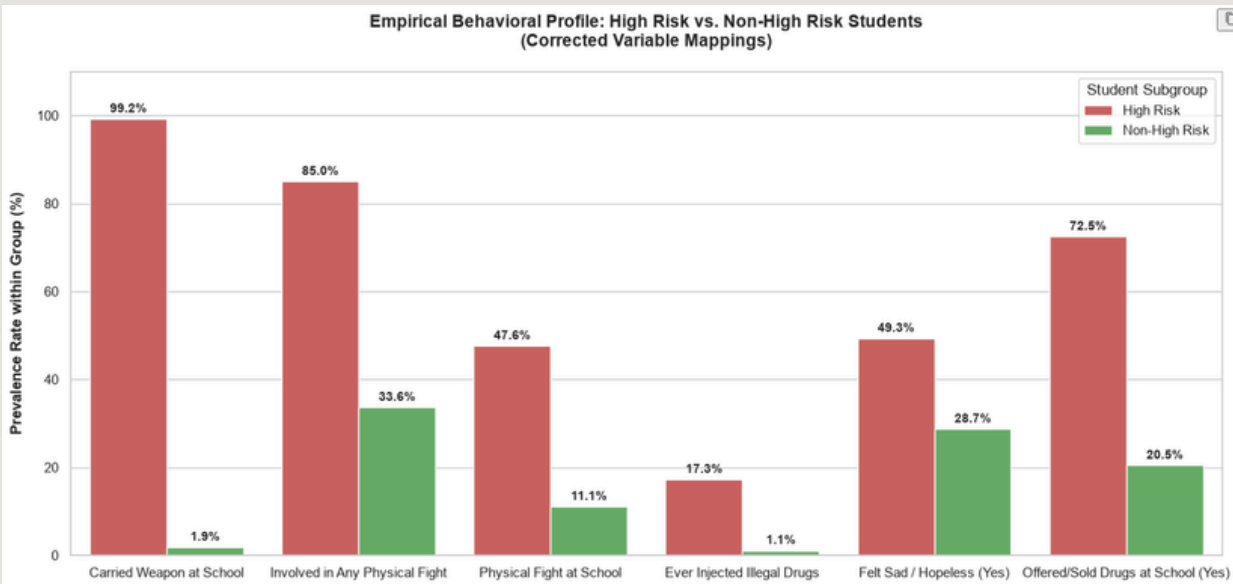
Result:

After Tukey HSD test the 9th grade(p=0.0156) and 10th grade(p=0.0192) are significant with 12th grade

And no other significant pairwise(p>0.05)

EXTEND

Analyze what common characteristics exist in high-risk groups



CONCLUSION

Compared to older grades, there are more high-risk students in the lower grades, and half of the students are in the sadness group. So, besides controlling weapons and drugs, counseling interventions are also needed to achieve the best results.